

## 2.2.2 Algorithm vs Pseudocode

الگورتھم بتاتا ہے کہ کیا کرنا ہے، جبکہ سوڈو کوڈ (Pseudocode) بتاتا ہے کہ اسے کیسے کیا جا سکتا ہے۔

CS XI

Chapter 2: Computational Thinking & Algorithm

Lecture 03 of 13

★ Golden

## 2.2.2 Algorithm vs Pseudocode ★ Golden

### LEARN IT

Both algorithms and pseudocode are used in problem-solving, but they serve different purposes in the design phase. The **algorithm** says *what* must be done. The **pseudocode** shows *how* it could be implemented, using programming-like keywords such as IF, FOR and WHILE, but it still cannot be run on a computer.

| Point        | Algorithm                                                              | Pseudocode                                                                      |
|--------------|------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Definition   | A step-by-step logical procedure to solve a problem.                   | A human-readable representation of an algorithm using plain English & keywords. |
| Nature       | Conceptual and language-independent.                                   | Semi-formal and structured, but not executable code.                            |
| Purpose      | To define <b>what</b> needs to be done logically.                      | To present <b>how</b> it can be implemented in a readable format.               |
| Form         | Usually described in plain steps, bullets or numbered points.          | Written using structured constructs (IF, FOR, WHILE) like programming.          |
| Focus        | Focuses on the logic and steps of the problem.                         | Focuses on representing steps clearly before coding.                            |
| Audience     | For anyone trying to understand the logic (including non-programmers). | Mainly for programmers or students preparing for coding.                        |
| Format rules | No fixed format; can be informal.                                      | Follows standard pseudocode conventions for clarity.                            |

### WORKED EXAMPLE

The same problem written both ways: print whether a number is even or odd.

#### Algorithm

Step 1: Start  
Step 2: Read a number N  
Step 3: Divide N by 2 and find the remainder  
Step 4: If the remainder is 0, print "Even", otherwise print "Odd"  
Step 5: End

#### Pseudocode

```
START
INPUT N
SET R TO N MOD 2
IF R = 0 THEN
    PRINT "Even"
ELSE
    PRINT "Odd"
END IF
END
```

The equivalent Python program (real, executable code):

```
n = 7
if n % 2 == 0:
    print("Even")
else:
    print("Odd")
```

OUTPUT  
Odd

Problem → Algorithm (what) → Pseudocode (how) → Program (code) → Output

### REMEMBER IT

Teacher's analogy: an **algorithm** is like a recipe ("Bake a cake at 350 degrees"). **Pseudocode** is like a structured kitchen manual using standard verbs (SET temperature TO 350; WHILE cake is not baked, WAIT). The program is the oven actually doing it.

### COMMON MISTAKE

Pseudocode is **not** executable. If you type pseudocode into Python, it will give errors. Also, do not mix exact Python syntax (colons, brackets) into pseudocode answers unless asked — use keywords like IF ... THEN ... END IF.

اردو میں سمجھیں

الگورتھم بتاتا ہے کہ کیا کرنا ہے، جبکہ سوڈو کوڈ (Pseudocode) بتاتا ہے کہ اسے کیسے کیا جا سکتا ہے۔ الگورتھم سادہ مرحلوں میں لکھا جاتا ہے اور ہر کوئی اسے سمجھ سکتا ہے۔ سوڈو کوڈ میں، IF اور WHILE جیسے الفاظ استعمال ہوتے ہیں اور یہ زیادہ تر پروگرامرز کے لیے ہوتا ہے، لیکن اسے کمپیوٹر پر چلایا نہیں جا سکتا۔

### EXAM TIP

"Differentiate between algorithm and pseudocode" is a favourite 4–5 mark question. Draw a two-column table and give at least **five** points (definition, nature, purpose, form, audience). Adding the recipe vs kitchen-manual analogy and one small example of each earns full marks.

### CHECK YOURSELF

1. Which one — algorithm or pseudocode — uses keywords like FOR and WHILE?
2. Which one is meant for non-programmers too?
3. Can pseudocode be executed by a computer?

Answers: 1. Pseudocode. 2. Algorithm. 3. No; it is semi-formal and not executable.

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Next lecture: 2.3.1 Decomposition